Mechanism of Disease

Chronic Inflammation

SUMMARY OF MAIN POINTS OF ACUTE INFLAMMATION

- Rapid response of living tissue to any injury.
- Naked eye (Macroscopic): Redness, swelling, heat, pain & loss offunction.
- Microscopic: Vascular dilatation, exudate leaks into tissues , neutrophils emigrate.
- Changes controlled by many short-lived chemical mediators. Some can be manipulated by drugs.
- Neutrophils: Fast acting, short-lived phagocytes, engulf & degrade bacteria, dead tissue etc.
- Phagocytosis enhanced by opsonisation of particles, e.g. antibody or complement on surface.
- Bacterial killing largely oxygen dependent.
- Defects in the system lead to severe susceptibility to infection.

CHRONIC INFLAMMATION

- Chronic inflammation is a response of prolonged duration (weeks or months) in which inflammation, tissue injury, and attempts at repair coexist, in varying combinations.
- **How chronic inflammation occur?**

1.May 'take over' from acute inflammation

If damage is too severe to be resolved within a few days…

2. May arise de novo

Some autoimmune conditions (E.g. RA)

Some Chronic Infections (E.g. viral hepatitis)

"Chronic low-level irritation"

3. May develop alongside acute inflammation

In severe, persistent or repeated irritation

The effects of chronic inflammation

- **Fibrosis**

Gall bladder (chronic cholecystitis), chronic peptic ulcers,cirrhosis

- **Impaired function**

Chronic Inflammatory Bowel Disease

Rarely, increased function, e.g. mucus secretion, thyrotoxicosis

- **Atrophy**

Gastric mucosa, adrenal glands

- **Stimulation of immune response**

Macrophage-Lymphocyte interactions

Causes of Chronic Inflammation

1-Persistent infections

2-Hypersensitivity diseases

3-Prolonged exposure to potentially toxic agents, either , Exogenous or Endogenous.

1-Persistent infections

- persistent Infections by microorganisms that are difficult to eradicate, such as mycobacteria and certain viruses, fungi, and parasites.

These organisms often evoke an immune reaction called delayed-type hypersensitivity.

- In other cases, unresolved acute inflammation evolves into chronic inflammation, such as when an acute bacterial infection of the lung progresses to a chronic lung abscess.

2-Hypersensitivity diseases

- Chronic inflammation plays an important role in a group of diseases that are caused by excessive and inappropriate activation of the immune system .
- In autoimmune diseases, self (auto) antigens evoke a self-perpetuating immune reaction that results in chronic inflammation and tissue damage; examples of such diseases are rheumatoid arthritis and multiple sclerosis.
- In allergic diseases, chronic inflammation is the result of excessive immune responses against common environmental substances, as in **bronchial asthma.**

3-Prolonged exposure to potentially toxic agents, either
exogenous or endogenous

- An example of an exogenous agent is particulate silica, a non degradable inanimate material that, when inhaled for prolonged periods, results in an inflammatory lung disease called silicosis.
- Atherosclerosis is a chronic inflammatory process affecting the arterial wall that is thought to be induced, at least in part, by excessive production and tissue deposition of endogenous **cholesterol** and other lipids.

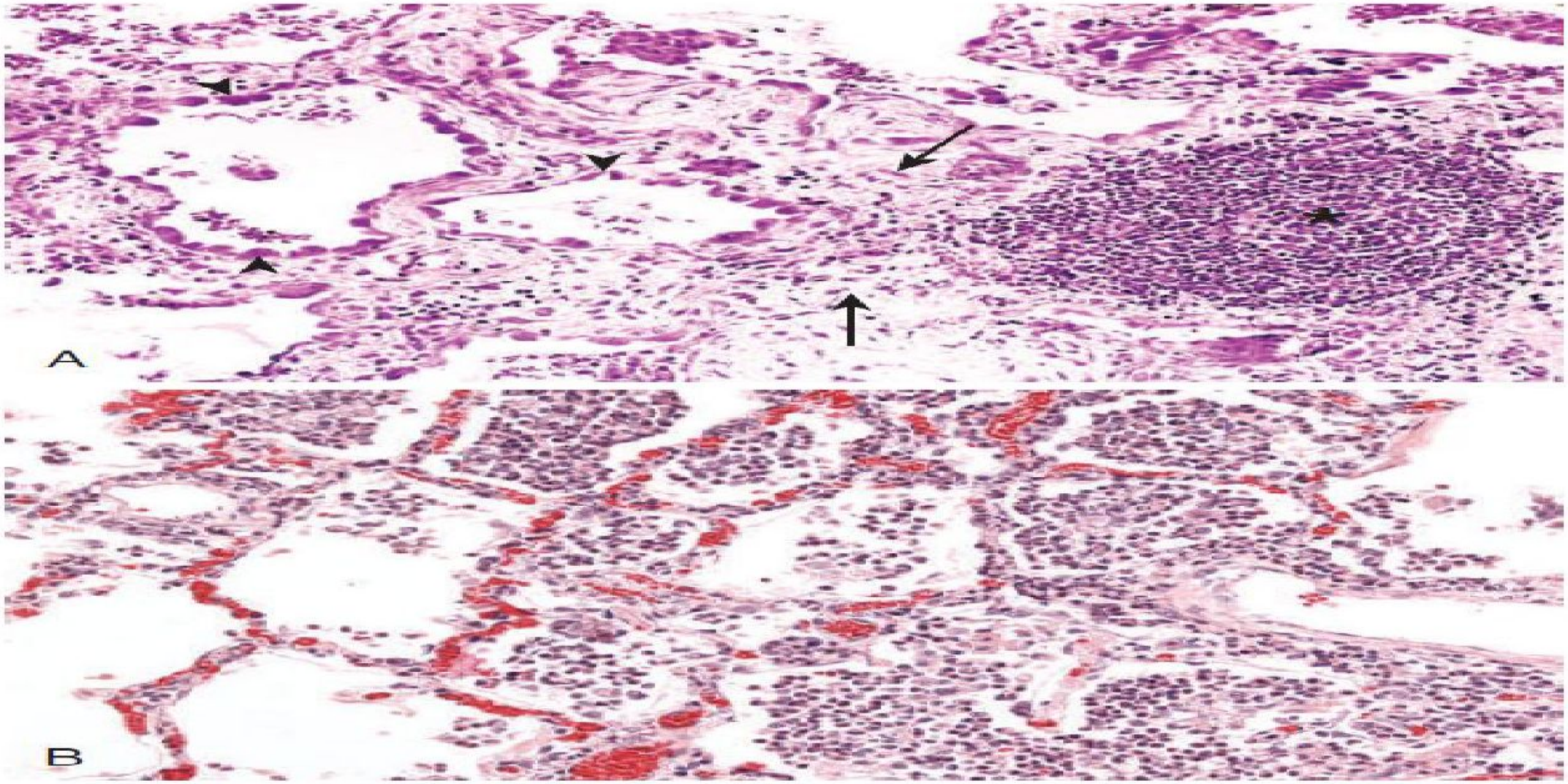


Fig. 3.17 (A) Chronic inflammation in the lung, showing all three characteristic histologic features: (1) collection of chronic inflammatory cells (*), (2) destruction of parenchyma (normal alveoli are replaced by spaces lined by cuboidal epithelium, *arrowheads*), and (3) replacement by connective tissue (fibrosis, *arrows*). (B) In contrast, in acute inflammation of the lung (acute bronchopneumonia), neutrophils fill the alveolar spaces and blood vessels are congested.

Morphologic Features of Chronic Inflammation

- In contrast to acute inflammation, which is manifested by vascular changes, edema, and predominantly neutrophilic infiltration, **chronic inflammation is characterized by the following:**
- Infiltration with mononuclear cells, which include macrophages, lymphocytes, and plasma cells.
- Tissue destruction, induced by the persistent offending agent or by the inflammatory cells.
- Attempts at healing by connective tissue replacement of damaged tissue, accomplished by angiogenesis (proliferation of small blood vessels) and, in particular, Fibrosis.

Cells and Mediators of Chronic Inflammation

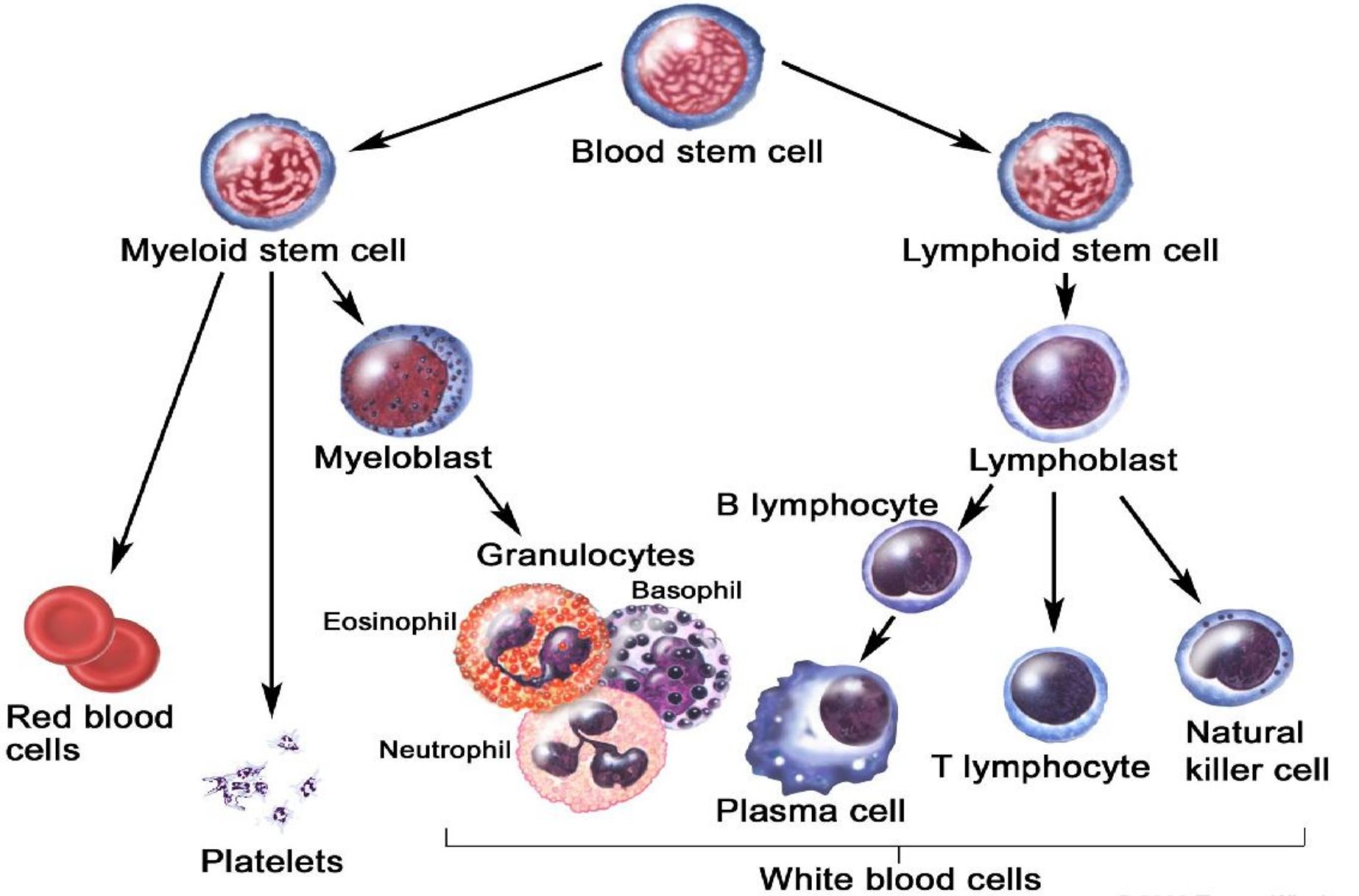
1-Macrophages

2-Lymphocytes

3-Mast Cells

4-Eosinophils

5-Neotrophils



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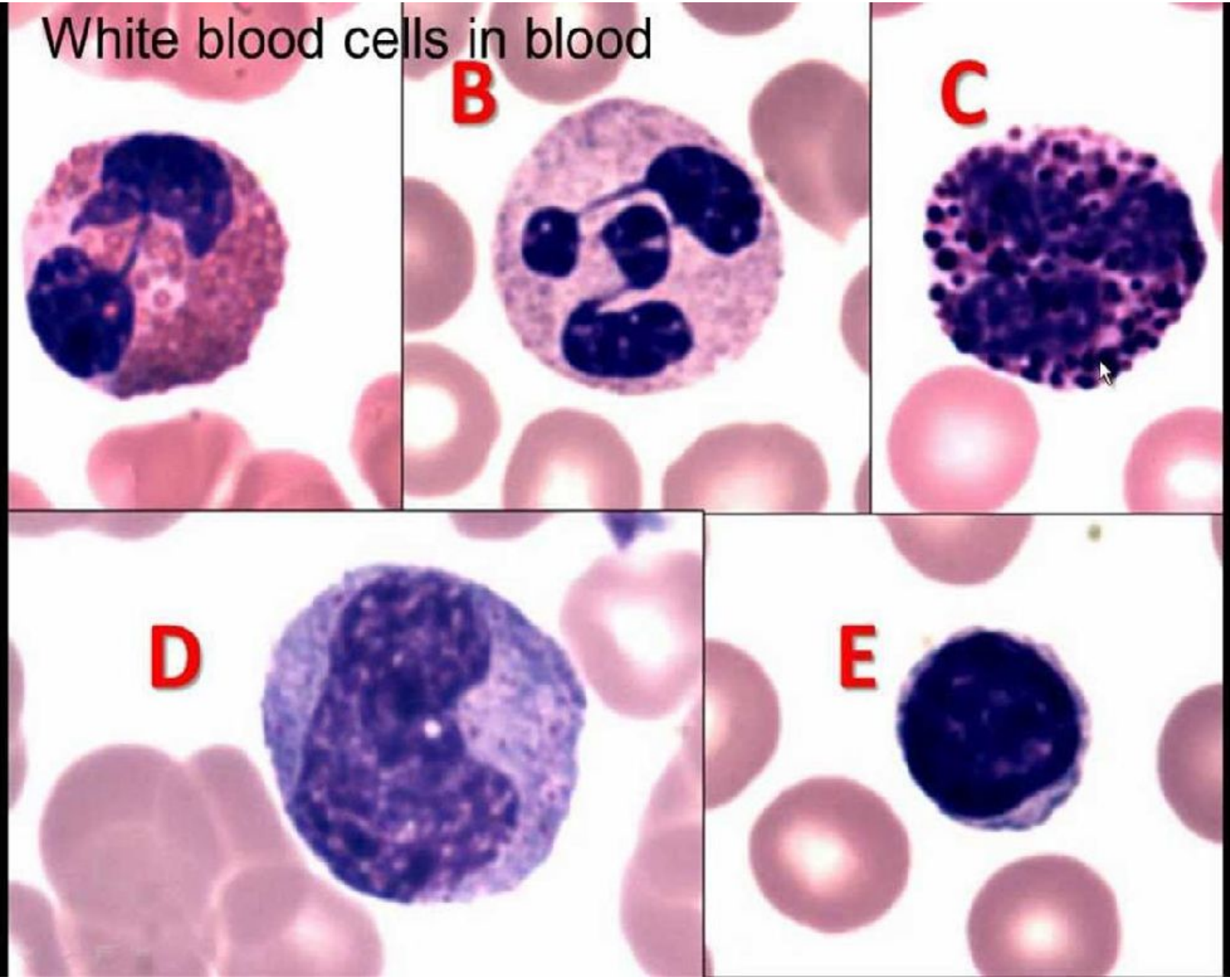
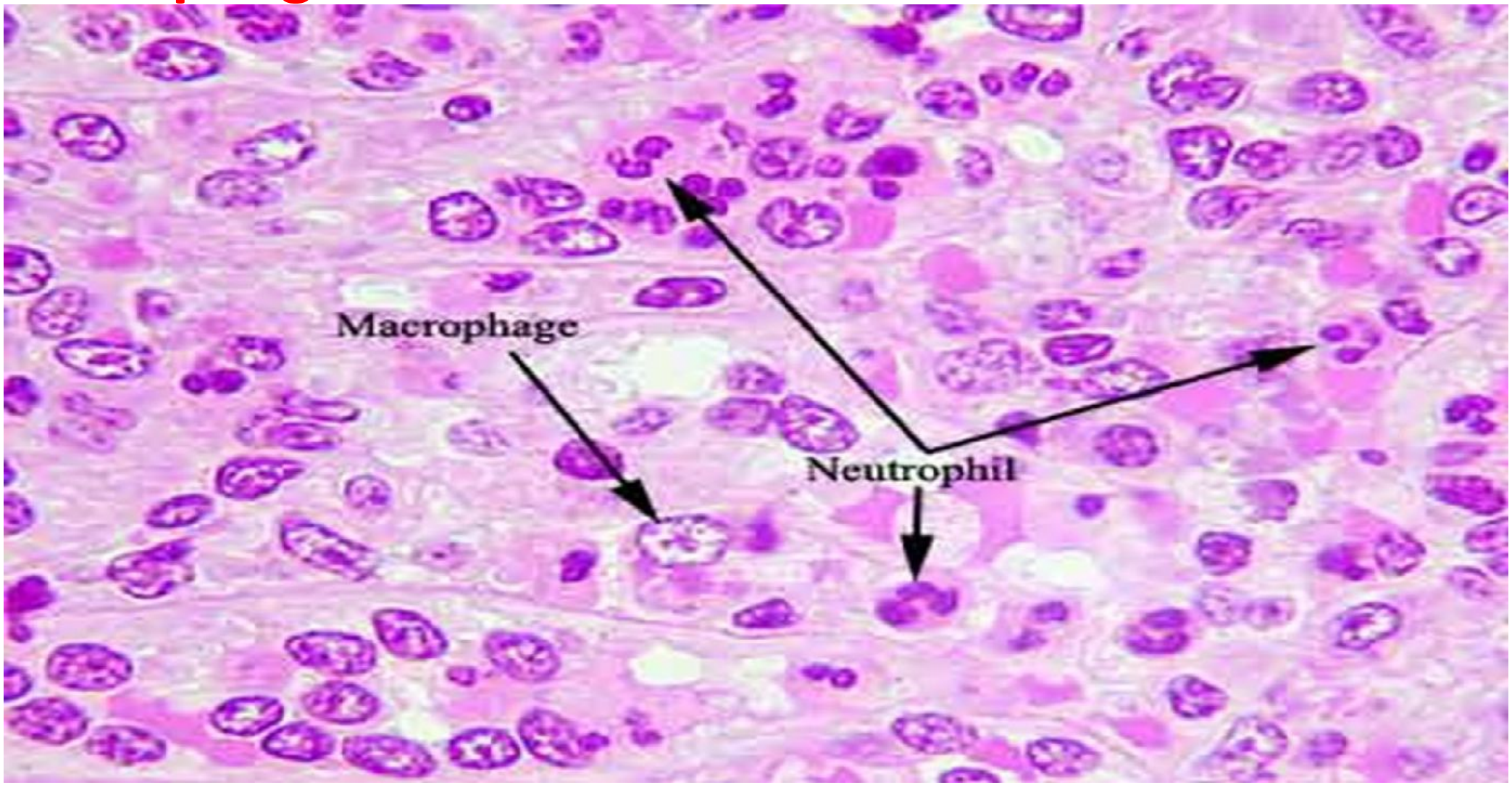

White blood cells in blood
B
C
D
E

Macrophages



Role of Macrophages

- The dominant cells in most chronic inflammatory reactions are macrophages .
- Macrophages are tissue cells derived from

1- hematopoietic stem cells in the bone marrow

2- progenitors in the embryonic yolk sac and fetal liver during early development

3- Circulating cells of this lineage are known as monocytes.

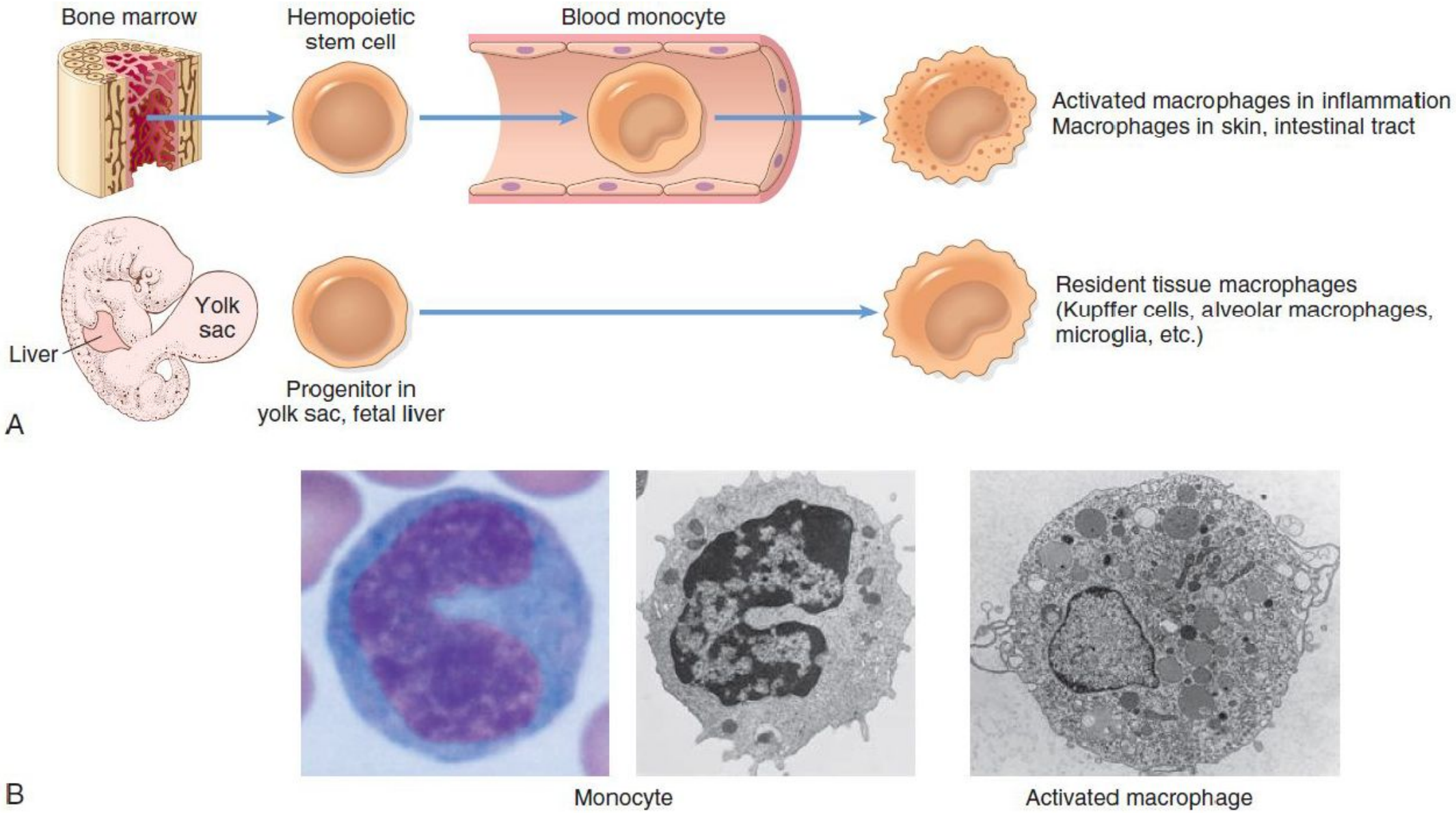


Fig. 3.18 Maturation of mononuclear phagocytes. (A) During inflammatory reactions, the majority of tissue macrophages are derived from hematopoietic precursors. Some long-lived resident tissue macrophages are derived from embryonic precursors that populate the tissues early in development. (B) The morphology of a monocyte and activated macrophage.

- Macrophages are normally diffusely scattered in most connective tissues.
- In addition, they found in specific locations in organs such as the liver (where they are called Kupffer cells), spleen and lymph nodes (sinus histiocytes), central nervous system (microglial cells), and lungs (alveolar macrophages).
- In inflammatory reactions, progenitors in the bone marrow give rise to monocytes, which enter the blood, migrate into various tissues, and differentiate into macrophages.
- Entry of blood monocytes into tissues is governed by the same factors that are involved in neutrophil emigration, such as adhesion molecules and chemokines..

- The half-life of blood monocytes is about 1 day, whereas the life span of
- tissue macrophages is several months or years.
- Thus, macrophages often become the dominant cell population in inflammatory reactions within 48 hours of onset.
- **Macrophages**

- Important in acute and chronic inflammation

- Various levels of activation

- Functions

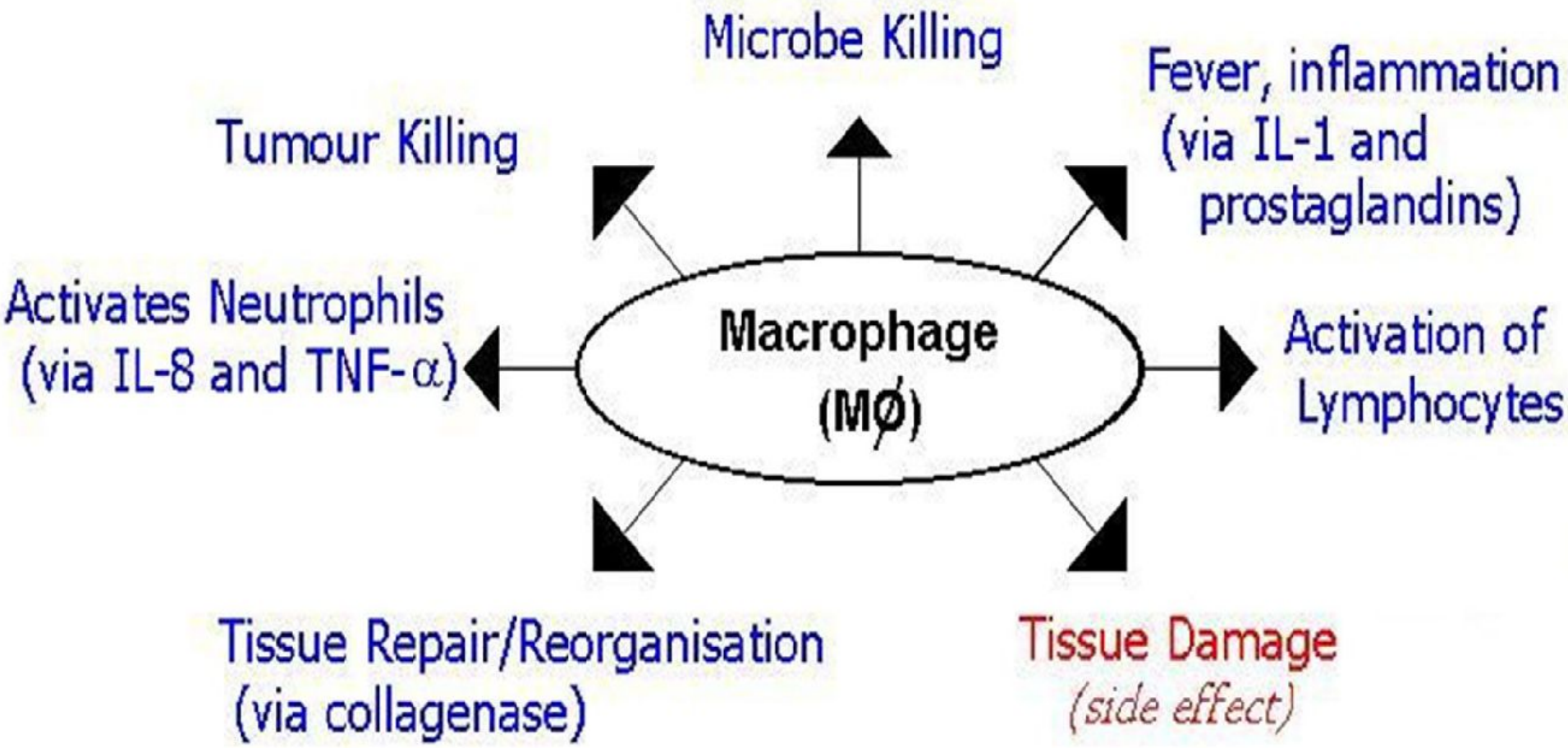
1. Phagocytosis and destruction of debris and bacteria

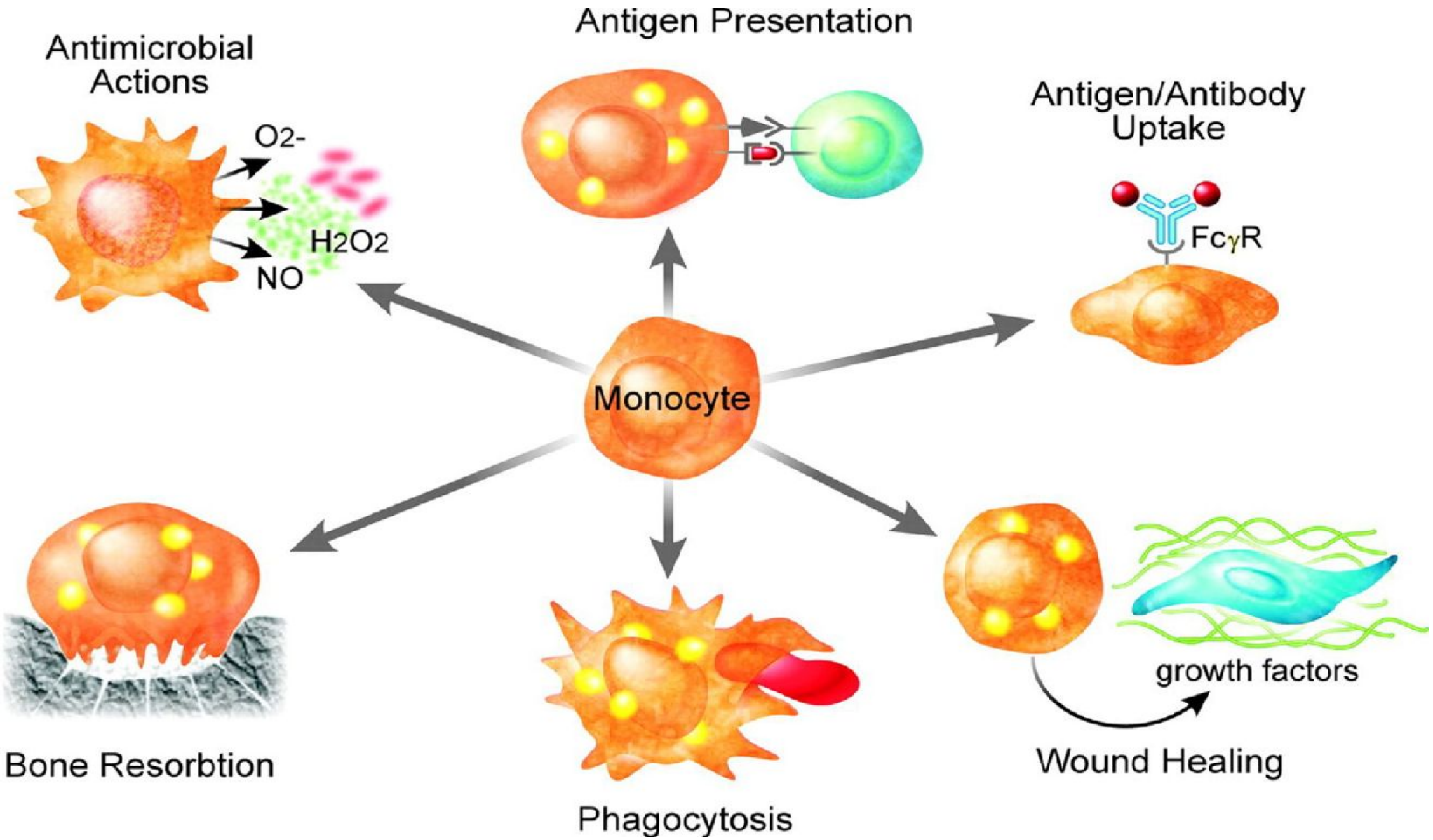
2.Processing and presentation of antigen to the immune system

3.Synthesis of cytokines, complement components, clotting factors and proteases

4. Control of other cells via cytokine release.

macrophages




Antimicrobial Actions
O2-
H2O2
NO
Antigen Presentation
Antigen/Antibody Uptake
FcγR
Monocyte
Bone Resorbtion
Phagocytosis
growth factors
Wound Healing

pathways of macrophage activation
1-classical pathway

- Classical macrophage activation may be induced bymicrobial products such as **endotoxin**, which engage TLRs and other sensors, and by **T cell–derived signals**, importantly the cytokine IFN-γ, in immune responses.
- Classically activated (also called M1) macrophages produce NO and ROS and upregulate lysosomal enzymes, all of which enhance their ability to kill ingested organisms, and secrete cytokines that stimulate inflammation.

2-Alternative pathway

- Alternative macrophage activation is induced by cytokines other than IFN-γ, such as IL-4 and IL-13, produced by T lymphocytes and other cells.
- These macrophages are not actively microbicidal; instead, the principal function of alternatively activated (M2) macrophages is in tissue repair.
- They secrete growth factors that promote angiogenesis, activate fibroblasts, and stimulate collage Synthesis.

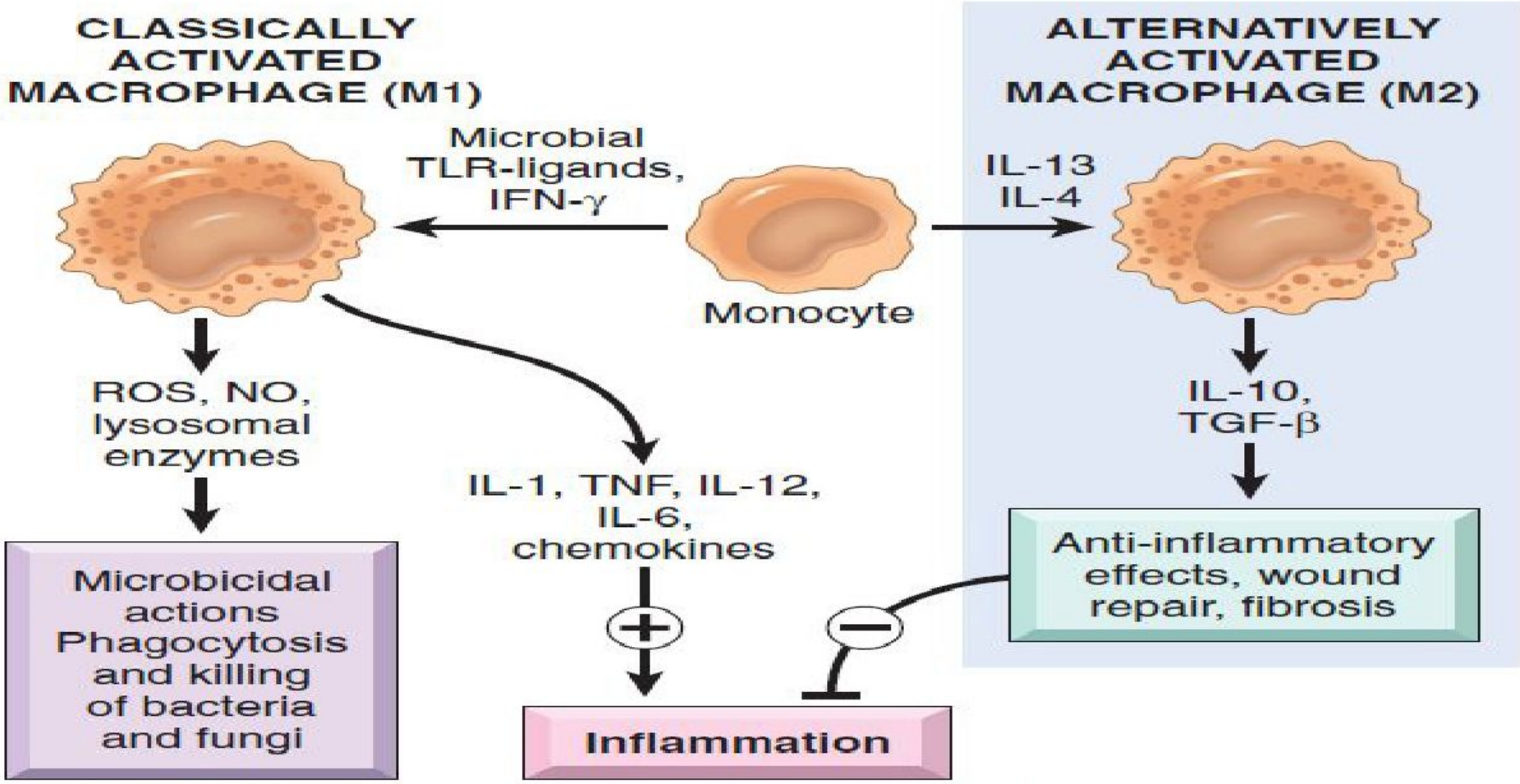
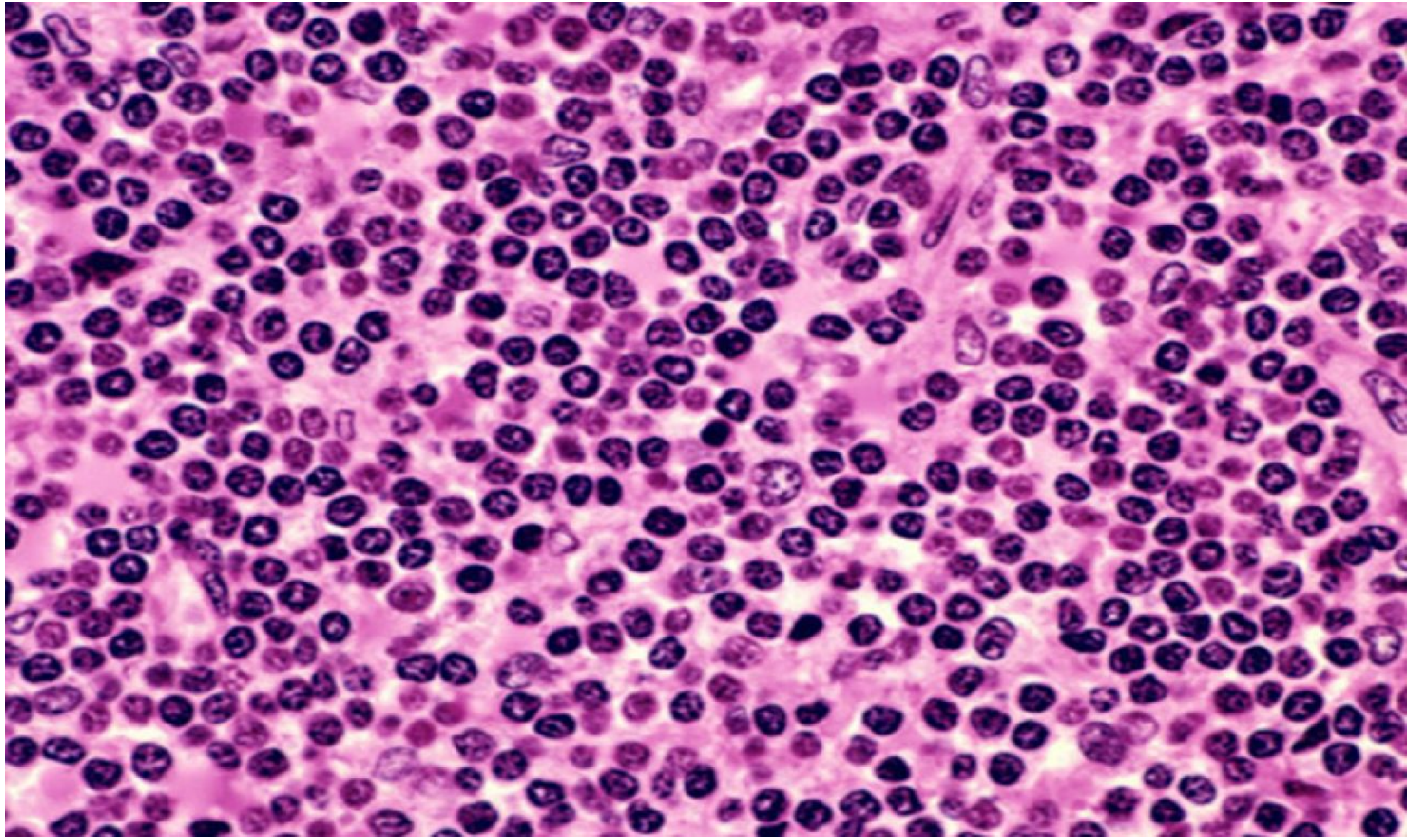


Fig. 3.19 Classical and alternative macrophage activation. Different stimuli activate monocytes/macrophages to develop into functionally distinct populations. Classically activated macrophages are induced by microbial products and cytokines, particularly IFN-γ. They phagocytose and destroy microbes and dead tissues and can potentiate inflammatory reactions. Alternatively activated macrophages are induced by other cytokines and are important in tissue repair and the resolution of inflammation.

Role of Lymphocytes

- Microbes and other environmental antigens activate T and B lymphocytes, which amplify and propagate chronic inflammation.
- **Sometimes called ‘chronic inflammatory cells’**
- these cells are often present in chronic inflammation and, when they are activated, the inflammation tends to be persistent and severe. Some of the strongest chronic inflammatory reactions, such as granulomatous inflammation, are dependent on lymphocyte responses.
- Lymphocytes may be the dominant population in the chronic inflammation seen in autoimmune and other hypersensitivity diseases



Lymphocytes

- There are three subsets of CD4+ T cells that secrete different cytokines and elicit different types of inflammation.
- TH1 cells produce the cytokine IFN-γ, which activates macrophages by the classical pathway.
- TH2 cells secrete IL-4, IL-5, and IL-13, which recruit and activate eosinophils and are responsible for the alternative pathway of macrophage activation.
- TH17 cells secrete IL-17 and other cytokines, which induce the secretion of chemokines responsible for recruiting neutrophils into the reaction.

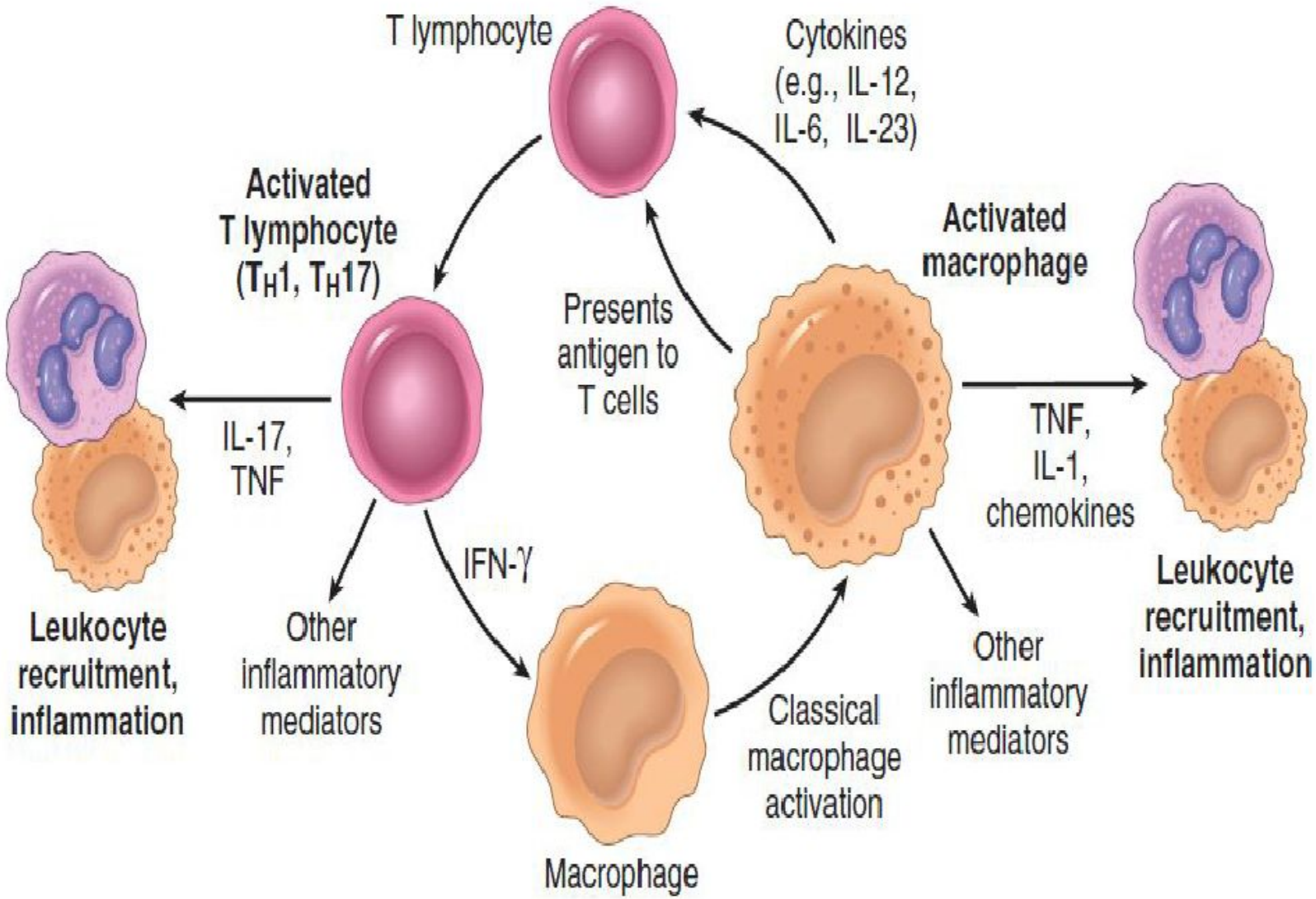
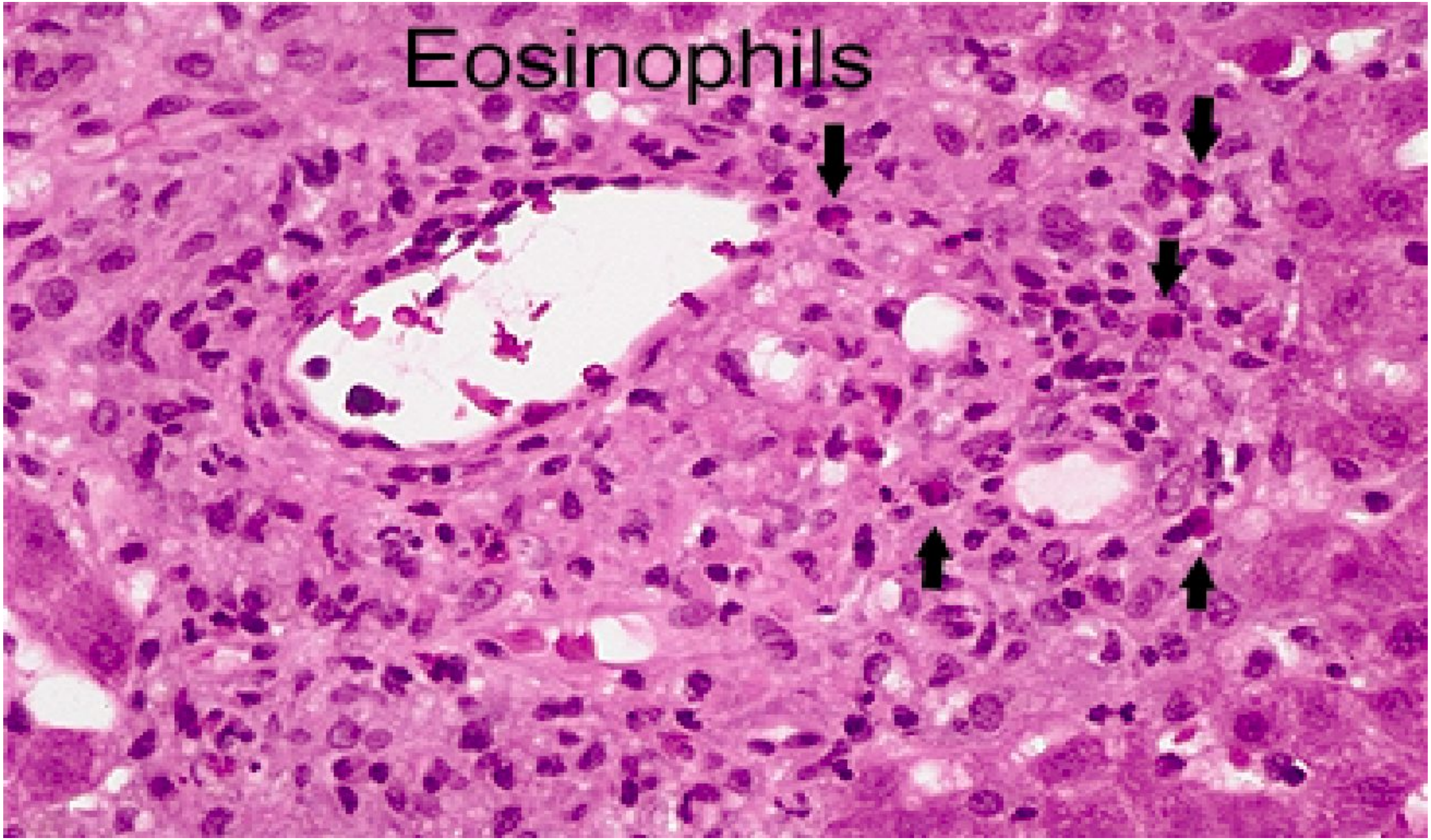


Fig. 3.20 Macrophage–lymphocyte interactions in chronic inflammation. Activated T cells produce cytokines that recruit macrophages (TNF, IL-17, chemokines) and others that activate macrophages (IFN-γ). Activated macrophages in turn stimulate T cells by presenting antigens and via cytokines such as IL-12.

Role of Eosinophils

- Eosinophils are abundant in immune reactions mediated by IgE and in **parasitic infections**.
- Their recruitment is driven by adhesion molecules similar to those used by neutrophils, and by specific chemokines (e.g., eotaxin) derived from leukocytes and epithelial cells.
- Eosinophils have granules that contain major basic protein, a highly cationic protein that is toxic to parasites but also injures host epithelial cells.
- This is why eosinophils are of benefit in controlling parasitic infections, yet also contribute to tissue damage in immune reactions such as **Allergies.**


Eosinophils

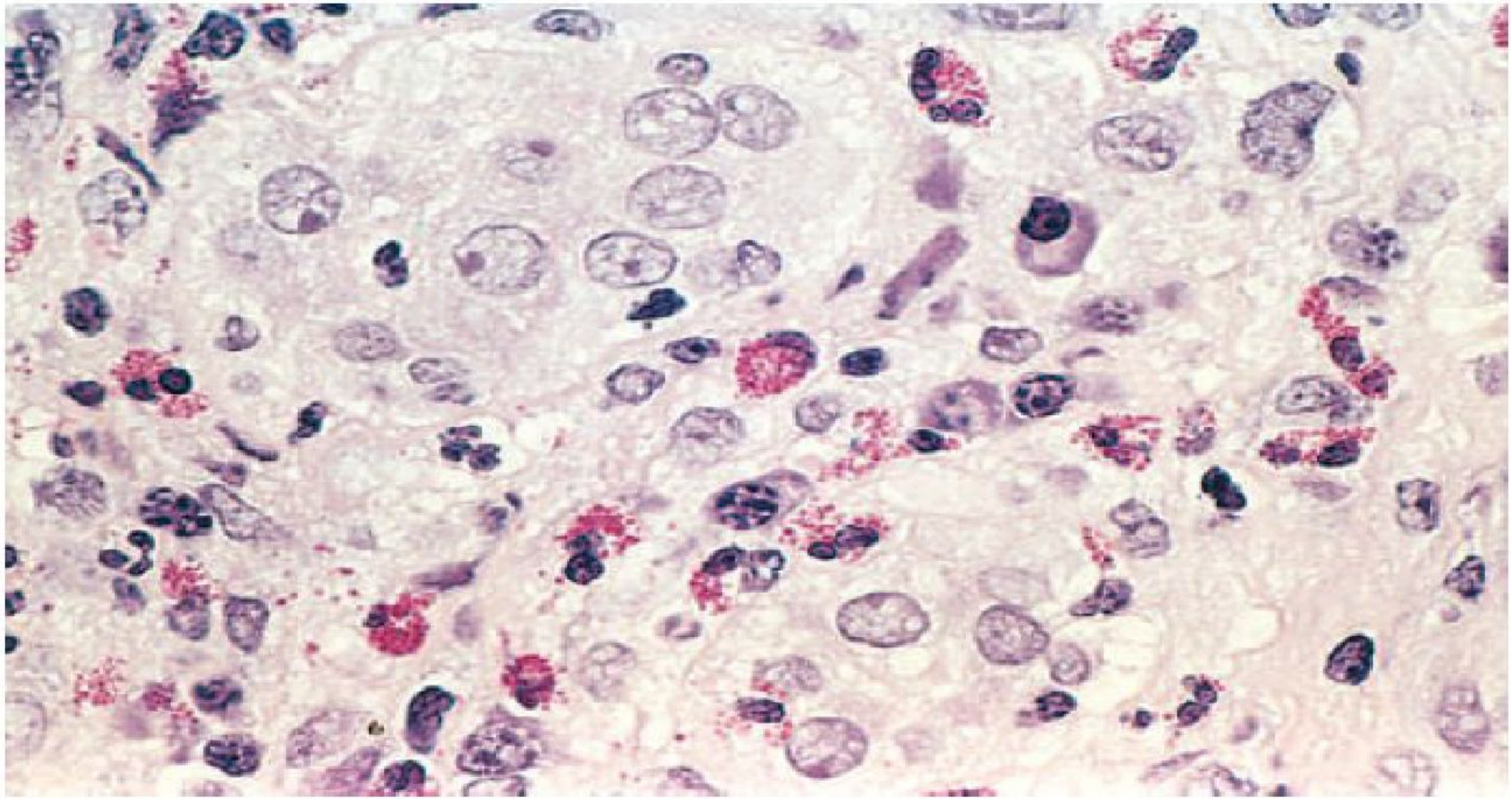


Fig. 3.21 A focus of inflammation containing numerous eosinophils.

Role of Mast Cells

- Mast cells arise from precursors in the bone marrow.
- They have many similarities with circulating basophils, but they do not arise from basophils, Mast cells (and basophils) express on their surface the receptor FcεRI, which binds the Fc portion of IgE antibody.
- In immediate hypersensitivity reactions ,IgE bound to the mast cells' Fc receptors specifically recognizes antigen, and in response the cells degranulate and release mediators, such as
- **histamine and prostaglandins**.
- This type of response occurs during allergic reactions to foods, insect venom, or drugs, sometimes with catastrophic results (e.g., **anaphylactic shock).**

Role of Neutrophils

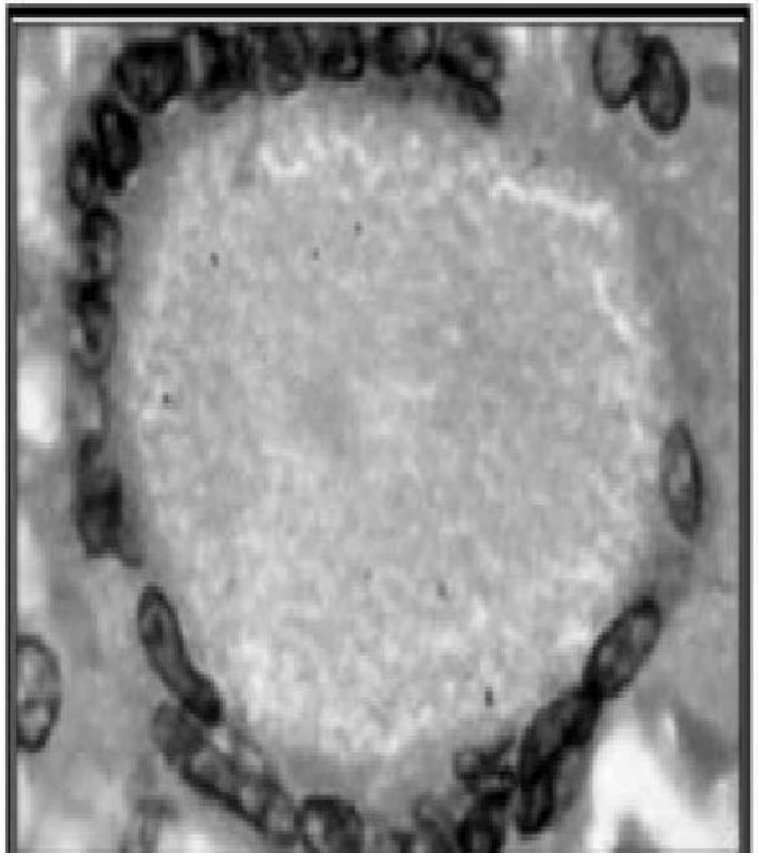
- Although neutrophils are characteristic of acute inflammation, many forms of chronic inflammation, lasting for months, continue to show large numbers of neutrophils, induced either by persistent microbes or by cytokines and other mediators produced by activated macrophages and T lymphocytes.
- In chronic bacterial infection of bone (osteomyelitis), a neutrophilic exudate can persist for many months.
- Neutrophils also are important in the chronic damage induced in lungs by smoking and other irritant stimuli.

Granulomatous Inflammation

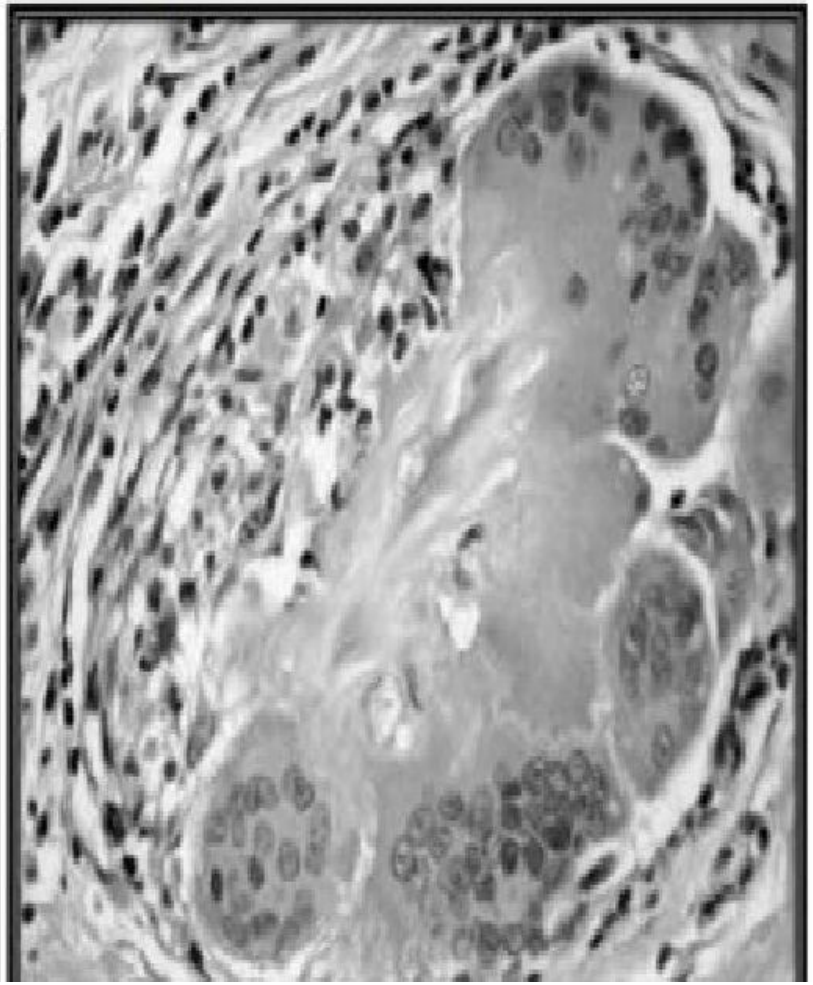
- Granulomatous inflammation is a form of chronic inflammation With Granuloma.
- **What is Granuloma?**
- collections of activated macrophages, often with T lymphocytes, and sometimes associated with central necrosis.
- The activated macrophages may develop abundant cytoplasm and begin to resemble epithelial cells, and are called epithelioid cells.
- Some activated macrophages may fuse, forming multinucleate giant cells.
- Granuloma formation is a cellular attempt to contain an offending agent that is difficult to eradicate.
- • Giant cells are multinucleate cells made by the fusion of macrophages, through the process of frustrated phagocytosis.
- There are several types recognised:

- Langhans Tuberculosis
- Foreign Body Type
- Touton. Fat Necrosis

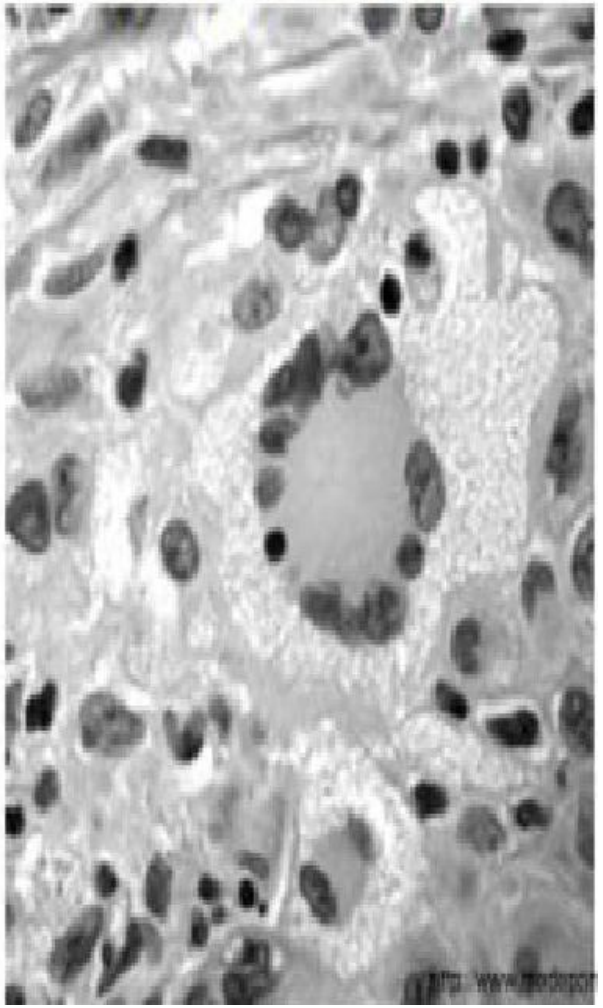
Langhans type giant cell - Tuberculosis



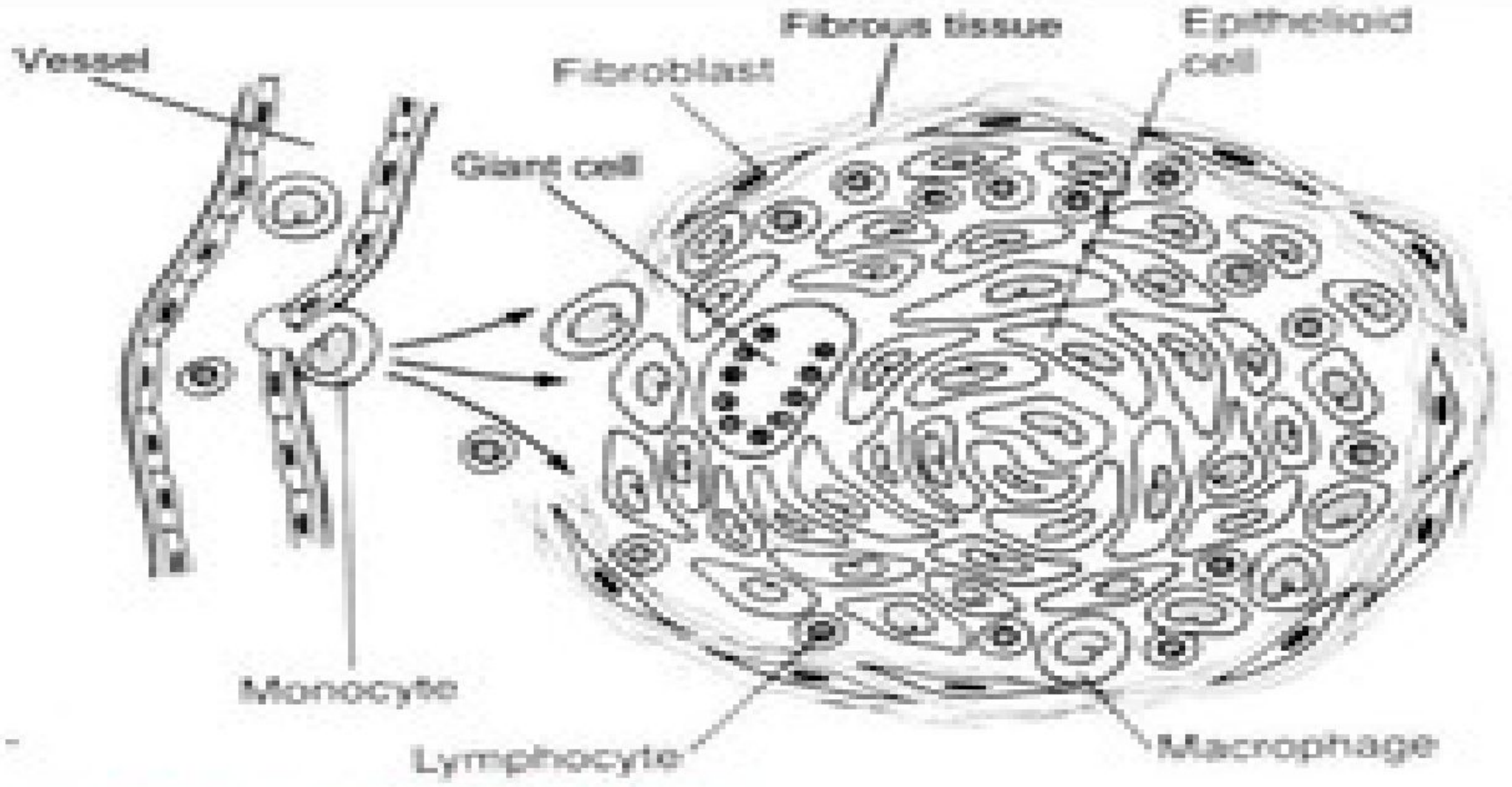
Foreign body type giant cells



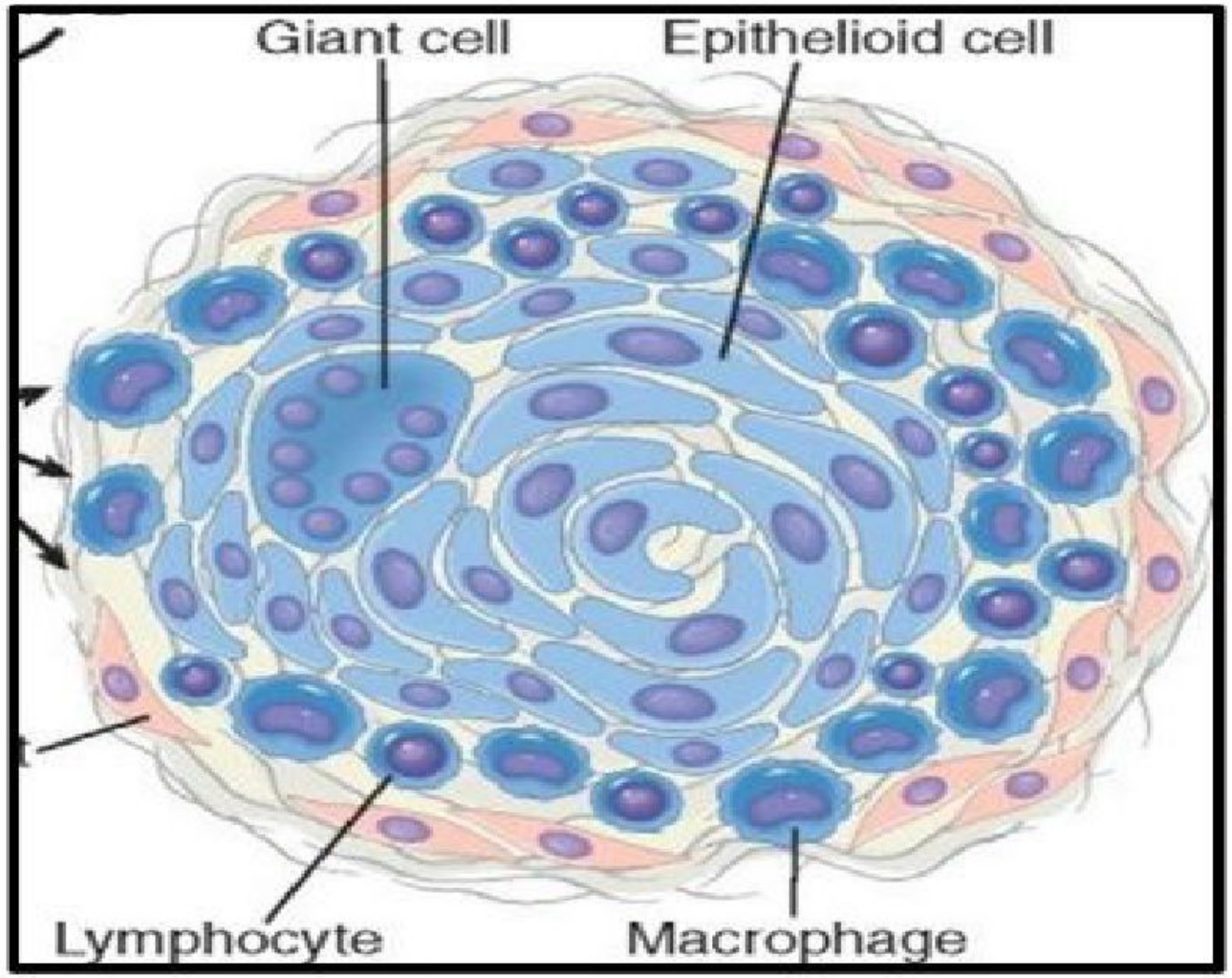
Touton Giant Cell



Formation of a granuloma



Taken from an early edition of Kumar "Basic Pathology"


Giant cell
Epithelioid cell
Lymphocyte
Macrophage

types of granulomas

Immune granulomas

- This type of immune response produces granulomas usually when the inciting agent cannot be readily eliminated, such as a persistent microbe or a self antigen.
- In such responses, macrophages activate T cells to produce cytokines such as IL-2, which activates other T cells, perpetuating the response, and IFN-γ, which activates themacrophages.

Foreign body granulomas

• Foreign body granulomas are seen in response to relatively • inert foreign bodies, in the absence of T cell–mediated immune responses.

Typically, foreign body granulomas form around materials such as talc (associated with intravenous drug abuse), sutures, or other fibers that are large enough to preclude phagocytosis by a macrophage but are not immunogenic

Table 3.9 Examples of Diseases With Granulomatous Inflammation

Disease	Cause	Tissue Reaction
Tuberculosis	*Mycobacterium tuberculosis*	Caseating granuloma (tubercle): focus of activated macrophages (epithelioid cells), rimmed by fibroblasts, lymphocytes, histiocytes, occasional Langhans giant cells; central necrosis with amorphous granular debris; acid-fast bacilli
Leprosy	*Mycobacterium leprae*	Acid-fast bacilli in macrophages; noncaseating granulomas
Syphilis	*Treponema pallidum*	Gumma: microscopic to grossly visible lesion, enclosing wall of macrophages; plasma cell infiltrate; central cells are necrotic without loss of cellular outline; organisms difficult to identify in tissue
Cat-scratch disease	Gram-negative bacillus	Rounded or stellate granuloma containing central granular debris and recognizable neutrophils; giant cells uncommon
Sarcoidosis	Unknown etiology	Noncaseating granulomas with abundant activated macrophages
Crohn disease (inflammatory bowel disease)	Immune reaction against undefined gut microbes and, possibly, self antigens	Occasional noncaseating granulomas in the wall of the intestine, with dense chronic inflammatory infiltrate

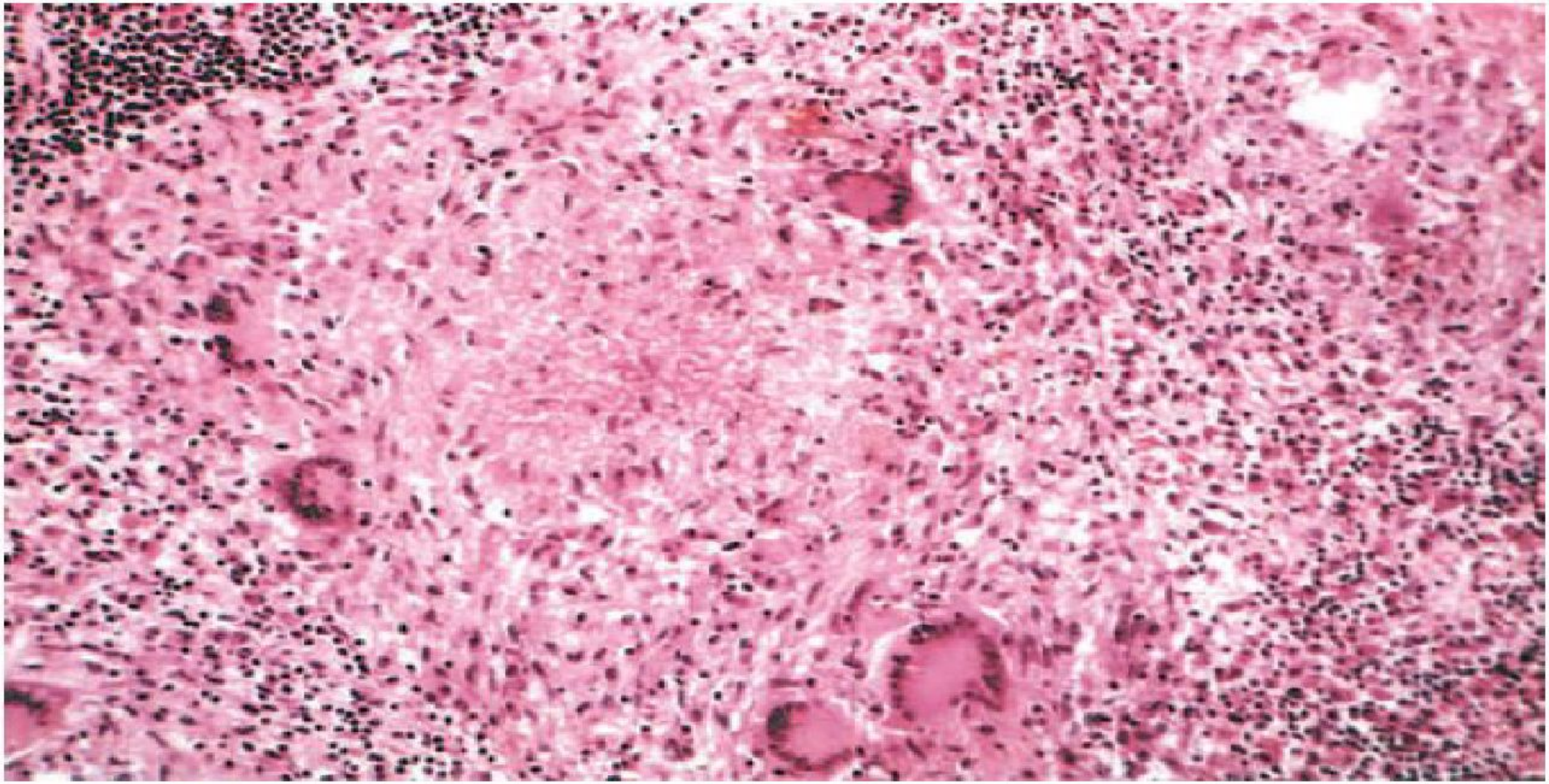


Fig. 3.22 Typical tuberculous granuloma showing an area of central necrosis surrounded by multiple multinucleate giant cells, epithelioid cells, and lymphocytes.

Some major clinical examples of chronic inflammation

- **Chronic Cholecystitis (Fibrosis)**
- Repeated obstruction of the gall bladder with gallstones.
- Repeated acute inflammation leads to chronic inflammation and fibrosis of the gall bladder wall.
- Treated with the surgical removal of the gall bladder.
- **Gastric Ulceration (Fibrosis)**
- Acute gastritis (alcohol, drugs)
- Chronic gastritis (Helicobacter pylori)
- Ulceration occurs because of an imbalance between acid production and mucosal defence.
- H. pylori triple treatment:

o PPI Inhibitor – E.g. Omeprazole

o 2 Antibiotics – E.g. Clarithromycin / Amoxicillin

- **Inflammatory Bowel Disease (Impaired Function)**
- Inflammatory disease affecting the large and small bowl. Patients present with diarrhoea, rectal bleeding and other symptoms.

- Ulcerative colitis

o Superficial

o Diarrhoea

o Bleeding

o Treat with immunosuppression, surgical ,removal of the large bowel (colectomy)

- **- Crohn's disease**

o Transmural

o Strictures

o Fistulae (Abnormal connection between two epithelia)

o Treat with lifestyle modifications, diet/hydration, immunosuppression

- **Liver Cirrhosis** (Fibrosis + Impaired Function)
- Chronic inflammation with fibrosis leading to disorganization of architecture and attempted regeneration.
- **Common causes of cirrhosis:**

- Alcohol
- Infection with HBV/HCV
- Immunological
- Fatty liver disease
- Drugs and toxins

- Liver cirrhosis cannot be reversed, so treatment involves lifestyle changes to prevent further damage, and transplantation of a new liver if necessary

- **Thyrotoxicosis** (Increased Function)

- Graves disease

- **Rheumatoid arthritis**

- Autoimmune disease

- Localised and systemic immune response

- Localised chronic inflammation leads to joint destruction

- Systemic immune response (Can affect other organs and cause amyloidosis)

Systemic Effects of Inflammation

- fever(pyrexia) is one of the most prominent manifestations of the acute-phase response, especially when inflammation is associated with infection.
- Substances that induce fever are called pyrogens.
- The increase in body temperature is caused by prostaglandins that are produced in the vascular and perivascular cells of the hypothalamus.
- Bacterial products, such as LPS (called exogenous pyrogens), stimulate leukocytes to release cytokines such as IL-1 and TNF (called endogenous pyrogens) that increase the enzymes (cyclooxygenases) that convert arachidonic acid into prostaglandins.
- In the hypothalamus, the prostaglandins , especially PGE2, stimulate the production of neurotransmitters that reset the temperature set point at a higher level.

Acute-phase proteins

- Acute-phase proteins are plasma proteins, mostly synthesized in the liver, Three of the best-known of these proteins are C-reactive protein (CRP), fibrinogen, and serum amyloid A (SAA) protein.
- Synthesis of these molecules in hepatocytes is stimulated by cytokines .
- Many acute-phase proteins, such as CRP and SAA, bind to microbial cell walls, and they may act as opsonins and fix complement.
- Fibrinogen binds to red cells and causes them to form stacks (rouleaux) that sediment more rapidly at unit gravity than do individual red cells.
- This is the basis for measuring the erythrocyte sedimentation rate as a simple test for an inflammatory response caused by any stimulus.

- Acute-phase proteins have beneficial effects during acute inflammation, but prolonged production of these proteins (especially SAA) in states of chronic inflammation can, in some cases, cause secondary Amyloidosis.
- Elevated serum levels of CRP serve as a marker for increased risk of myocardial infarction in patients with coronary artery disease.
- It is postulated that inflammation involving atherosclerotic plaques in the coronary arteries predisposes to thrombosis and subsequent Infarction.

Leukocytosis

- Leukocytosis is a common feature of inflammatory reactions, especially those induced by bacterial infections.
- The leukocyte count usually climbs to 15,000 or 20,000 cells/mL.
- The leukocytosis occurs initially because of accelerated release of cells from the bone marrow post mitotic reserve pool (caused by cytokines, including TNF and IL-1) and is therefore associated with a rise in the number of more immature neutrophils in the blood, referred to as a shift to the left.

- Other manifestations of the acute-phase response include increased heart rate and blood pressure decreased.
- sweating, mainly because of redirection of blood flow from cutaneous to deep vascular beds, to minimize heat loss through the skin; rigors (shivering), chills (search for warmth), anorexia, somnolence, and malaise, probably because of the actions of cytokines on brain cells.

- In severe bacterial infections (sepsis), the large amounts of bacteria and their products in the blood stimulate the production of enormous quantities of several cytokines, notably TNF and IL-1.
- High blood levels of cytokines cause widespread clinical and pathologic abnormalities such as disseminated intravascular coagulation, hypotensive shock, and metabolic disturbances including insulin resistance and hyperglycemia.
- A syndrome similar to septic shock may occur as a complication of noninfectious disorders, such as severe burns, trauma, pancreatitis, and other serious conditions. Collectively these reactions are called systemic inflammatory response syndrome (SIRS).

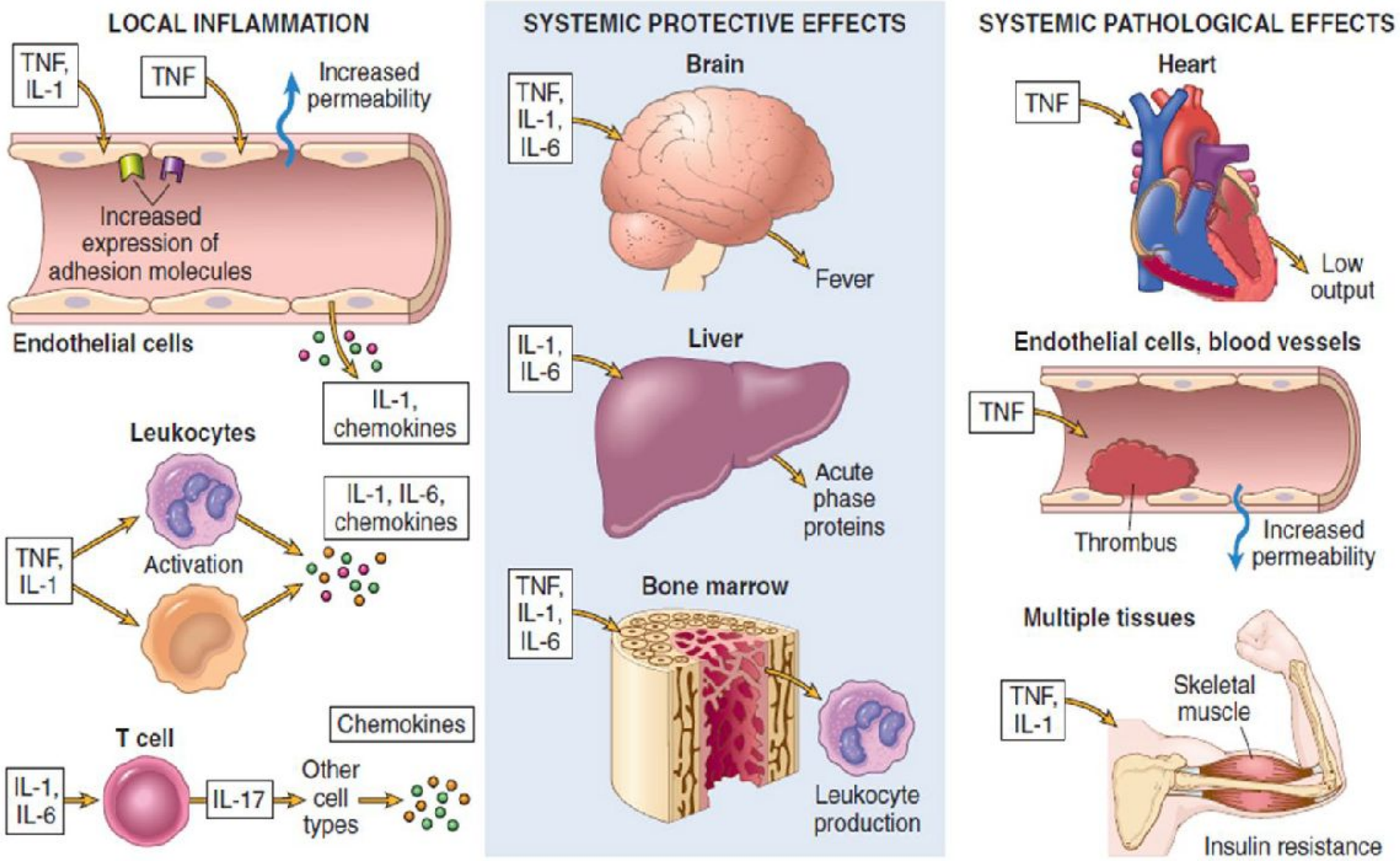


Fig. 3.10 Major roles of cytokines in acute inflammation. *PDGF*, Platelet-derived growth factor; *PGE*, prostaglandin E; *PGI*, prostaglandin I.

Thank you